

Comparitive Performance Report for tentacle-ec-2+2-16k-balance-off-rerun3 vs direct-reads-ec-2+2-16k-balance-off- rerun-pr2 vs direct-reads-ec-2+2-16k-balance-reads- on-alexpoc

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Comparison summary for tentacle-ec-2+2-16k-balance-off-rerun3 vs direct-reads-ec-2+2-16k-balance-off- rerun-pr2 vs direct-reads-ec-2+2-16k-balance-reads- on-alexpoc

Sequential Read	tentacle_ec_2+2_16k_balance_off_rerun3	direct_reads_ec_2+2_16k_balance_off_rerun_pr2	direct_reads_ec_2+2_16k_balance_reads_on_alexpoc
4K	76008@5.0ms	77727@4.9	178662@1.4
8K	75022@5.1ms	77329@5.0	177280@2.2
16K	72629@5.3ms	75168@5.1	165849@2.3
32K	54841@7.0ms	57447@6.7	82754@4.6
64K	3219@7.8ms	3353@7.5	4400@5.7
128K	5463@9.2ms	5711@8.8	6847@7.3
256K	7825@12.9ms	8330@12.1	9562@8.8
384K	8704@23.1ms	9469@21.3	10748@9.4
512K	9283@28.9ms	10213@26.3	11555@11.6
768K	9350@43.1ms	10354@38.9	11830@17.0
1024K	9496@42.4ms	10544@38.2	12648@21.2
2048K	8973@14.9ms	9725@27.6	12738@10.5
4096K	8490@31.6ms	8989@29.6	11733@11.4

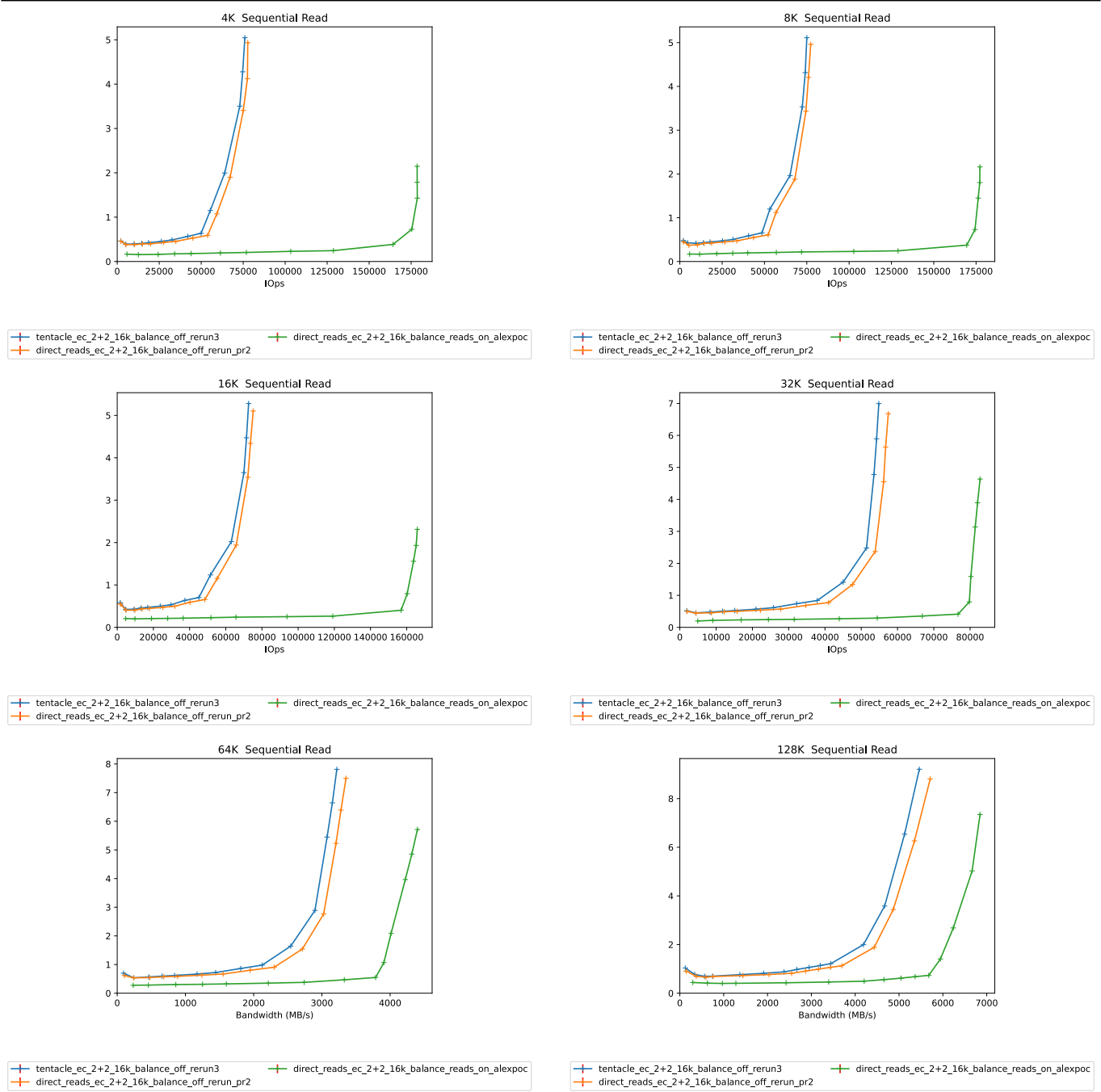
Random Read	tentacle_ec_2+2_16k_balance_off_rerun3	direct_reads_ec_2+2_16k_balance_off_rerun_pr2	direct_reads_ec_2+2_16k_balance_reads_on_alexpoc
4K	73319@5.2ms	75857@5.1	151700@2.5
8K	71092@5.4ms	73165@4.4	145011@2.6
16K	68554@5.6ms	70725@5.4	137160@2.8
32K	48537@7.9ms	49626@7.7	69392@5.5
64K	2811@8.9ms	2890@8.7	3752@6.7
128K	4789@10.5ms	4996@10.1	6155@5.4
256K	7312@9.2ms	7745@13.0	9107@9.2
384K	7964@6.3ms	8947@11.2	10568@5.9
512K	8703@30.8ms	9687@27.7	11768@5.7
768K	8941@45.0ms	9963@40.5	12280@16.4
1024K	9146@44.0ms	10142@39.7	12976@10.3
2048K	8839@30.4ms	9438@28.4	12702@10.5
4096K	8447@31.8ms	8985@29.8	11830@11.3

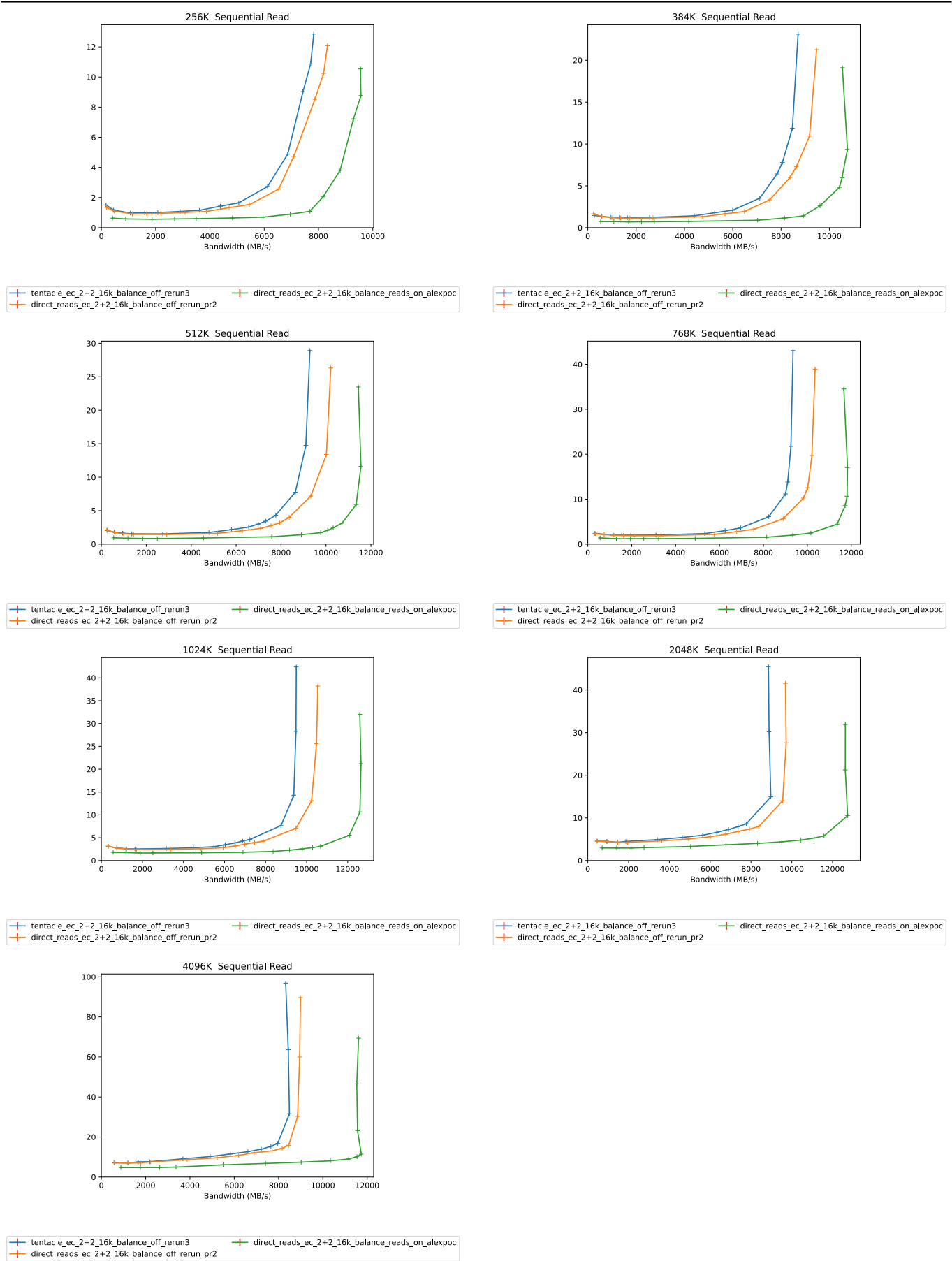
Random Read/Write	tentacle_ec_2+2_16k_balance_off_rerun3	direct_reads_ec_2+2_16k_balance_off_rerun_pr2	direct_reads_ec_2+2_16k_balance_reads_on_alexpoc
4K_70/30	36665@10.5ms	37441@8.5	48221@8.0
16K_70/30	34355@11.2ms	35734@10.7	45386@7.0

Random					
Read/Write	tentacle_ec_2+2_16k_balance_off_rerun3	direct-reads-ec-2+2-16k-balance-off-rerun-pr2	direct-reads-ec-2+2-16k-balance-reads-on-alexpoc	tentacle_ec_2+2_16k_balance_off_rerun3	direct-reads-ec-2+2-16k-balance-off-rerun-pr2
64K_70/30	1699@14.8ms	1766@11.9	4%	2026@8.3	19%
64K_30/70	1221@13.7ms	1274@19.7	4%	1369@12.2	12%

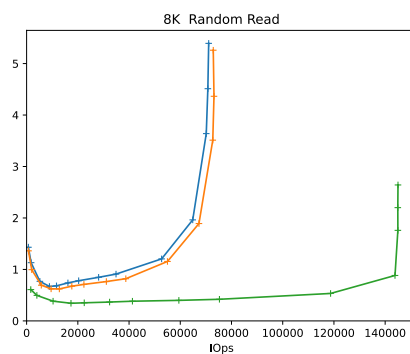
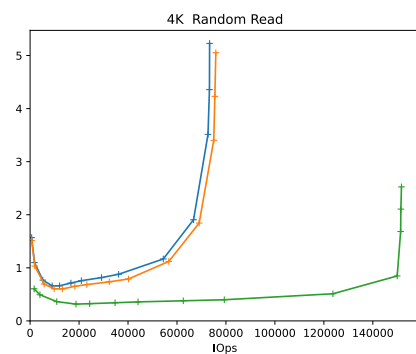
Response Curves

Sequential Read



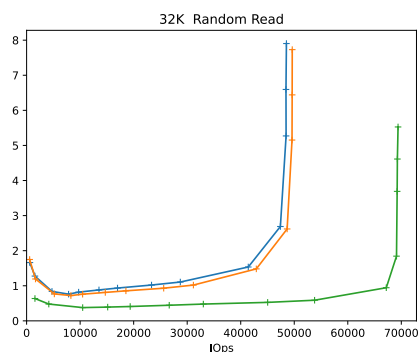
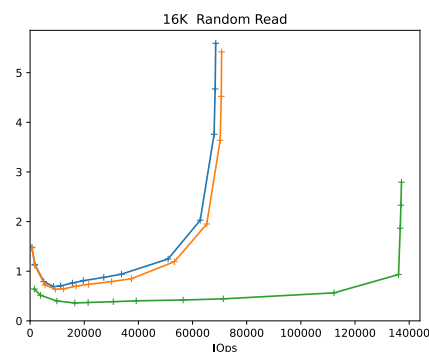


Random Read



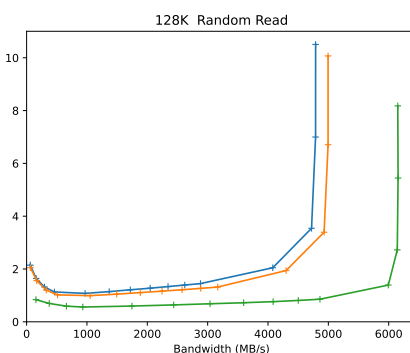
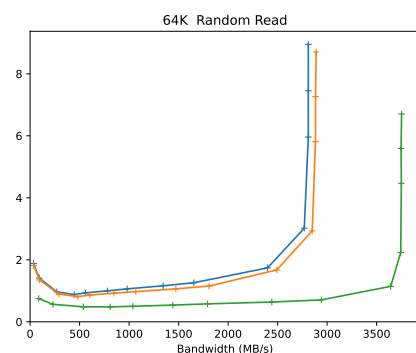
tentacle_ec_2+2_16k_balance_off_rerun3 direct_reads_ec_2+2_16k_balance_reads_on_alex poc
direct_reads_ec_2+2_16k_balance_off_rerun_pr2

tentacle_ec_2+2_16k_balance_off_rerun3 direct_reads_ec_2+2_16k_balance_reads_on_alex poc
direct_reads_ec_2+2_16k_balance_off_rerun_pr2



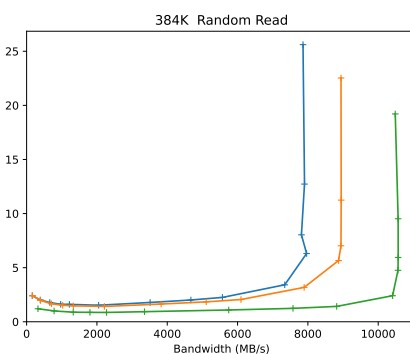
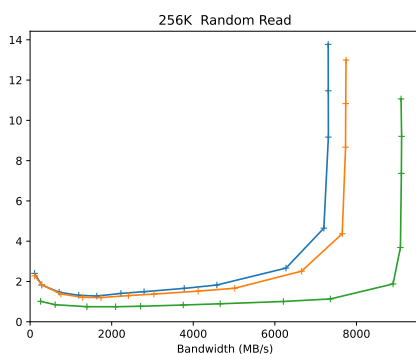
tentacle_ec_2+2_16k_balance_off_rerun3 direct_reads_ec_2+2_16k_balance_reads_on_alex poc
direct_reads_ec_2+2_16k_balance_off_rerun_pr2

tentacle_ec_2+2_16k_balance_off_rerun3 direct_reads_ec_2+2_16k_balance_reads_on_alex poc
direct_reads_ec_2+2_16k_balance_off_rerun_pr2



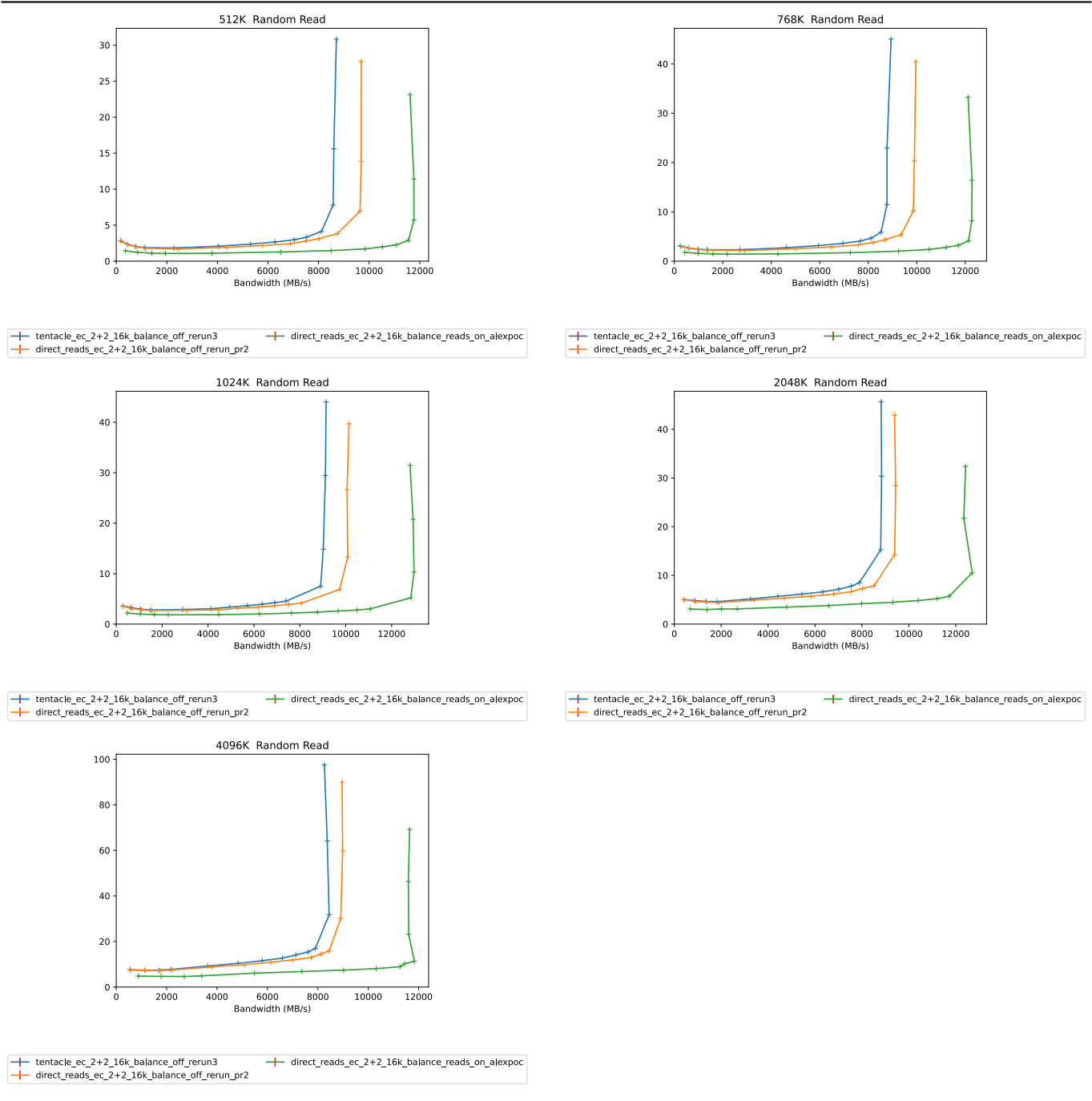
tentacle_ec_2+2_16k_balance_off_rerun3 direct_reads_ec_2+2_16k_balance_reads_on_alex poc
direct_reads_ec_2+2_16k_balance_off_rerun_pr2

tentacle_ec_2+2_16k_balance_off_rerun3 direct_reads_ec_2+2_16k_balance_reads_on_alex poc
direct_reads_ec_2+2_16k_balance_off_rerun_pr2

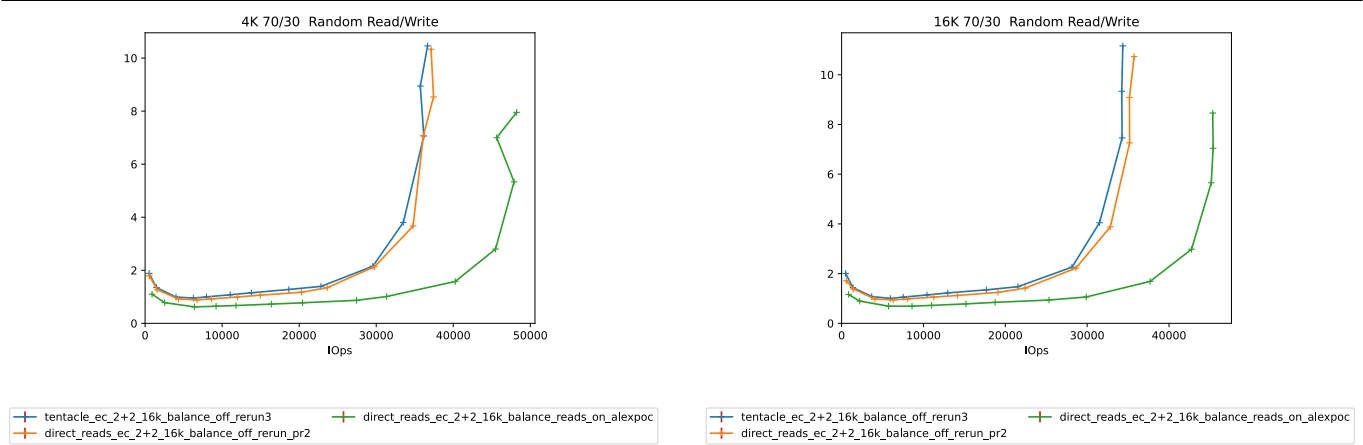


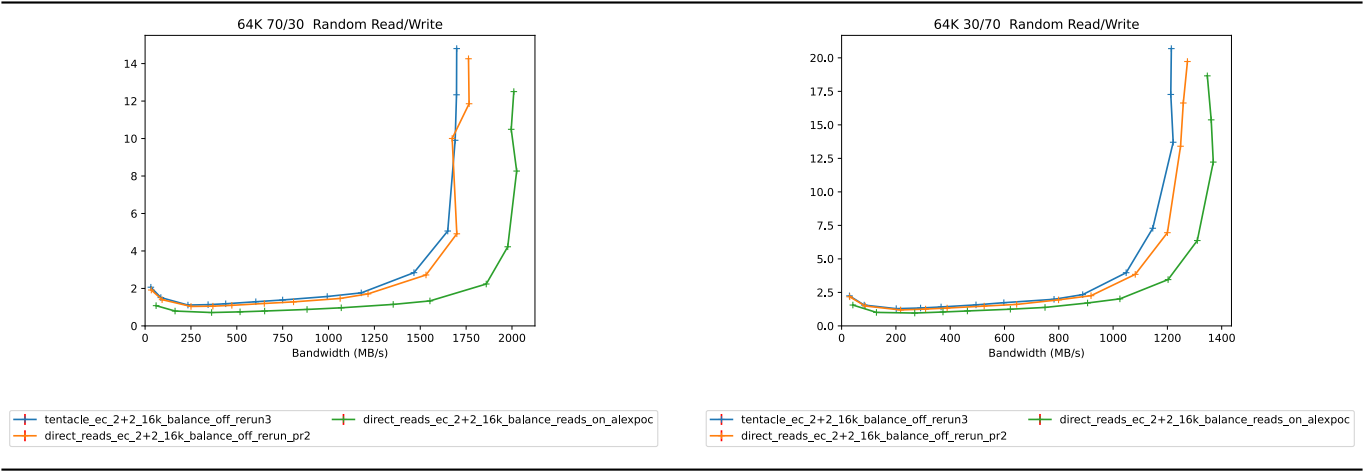
tentacle_ec_2+2_16k_balance_off_rerun3 direct_reads_ec_2+2_16k_balance_reads_on_alex poc
direct_reads_ec_2+2_16k_balance_off_rerun_pr2

tentacle_ec_2+2_16k_balance_off_rerun3 direct_reads_ec_2+2_16k_balance_reads_on_alex poc
direct_reads_ec_2+2_16k_balance_off_rerun_pr2



Random Read/Write





Configuration yaml files

Only yaml files that differ by more than 20 lines from the yaml file for the baseline directory will be added here in addition to the baseline yaml

results

```

librbd fio:
  cmd_path: /usr/local/bin/fio2
  create_report: true
  fio_out_format: json
  log_avg_msec: 100
  log_bw: true
  log_iops: true
  log_lat: true
  norandommap: true
  osd_ra:
    - 4096
  poolname: rbd_replicated
  prefill:
    blocksize: 64k
    numjobs: 1
  procs_per_volume:
    - 1
  ramp: 30
  rbdname: cbt-librbd fio
  time: 90
  time_based: true
  use_existing_volumes: true
  vol_size: 1000
  volumes_per_client:
    - 16
  wait_pgautoscaler_timeout: 10
  workloads:
    128k randomread:
      jobname: randread
      mode: randread
      numjobs:
        - 1
      op_size: 131072
      total_iodepth:
        - 1
        - 2
        - 3
        - 4
        - 8
        - 12
        - 16
        - 20
        - 24
        - 28
        - 32
        - 64
        - 128
        - 256
        - 384
    128k sequentialread:
      jobname: seqread
      mode: read
      numjobs:
        - 1
      op_size: 131072
      total_iodepth:
        - 1
        - 2
        - 3
        - 4
        - 8

```

```

- 12
- 16
- 20
- 24
- 28
- 32
- 64
- 128
- 256
- 384
16kmixread70:
  jobname: mixread
  mode: randrw
  numjobs:
    - 1
  op_size: 16384
  rwmixread: 70
  total_iodepth:
    - 1
    - 2
    - 4
    - 6
    - 8
    - 12
    - 16
    - 24
    - 32
    - 64
    - 128
    - 256
    - 320
    - 384
16krandomread:
  jobname: randread
  mode: randread
  numjobs:
    - 1
  op_size: 16384
  total_iodepth:
    - 1
    - 2
    - 4
    - 6
    - 8
    - 12
    - 16
    - 24
    - 32
    - 64
    - 128
    - 256
    - 320
    - 384
16ksequentialread:
  jobname: seqread
  mode: read
  numjobs:
    - 1
  op_size: 16384
  total_iodepth:
    - 1
    - 2
    - 4

```

```

- 6
- 8
- 12
- 16
- 24
- 32
- 64
- 128
- 256
- 320
- 384
1Mrandomread:
  jobname: randread
  mode: randread
  numjobs:
    - 1
  op_size: 1048576
  total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 8
    - 12
    - 16
    - 20
    - 24
    - 28
    - 32
    - 64
    - 128
    - 256
    - 384
1Msequentialread:
  jobname: seqread
  mode: read
  numjobs:
    - 1
  op_size: 1048576
  total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 8
    - 12
    - 16
    - 20
    - 24
    - 28
    - 32
    - 64
    - 128
    - 256
    - 384
256krandomread:
  jobname: randomread
  mode: randread
  numjobs:
    - 1
  op_size: 262144
  total_iodepth:
    - 1

```

```

- 2
- 4
- 6
- 8
- 12
- 16
- 24
- 32
- 64
- 128
- 256
- 320
- 384
256ksequentialread:
  jobname: seqread
  mode: read
  numjobs:
    - 1
  op_size: 262144
  total_iodepth:
    - 1
    - 2
    - 4
    - 6
    - 8
    - 12
    - 16
    - 24
    - 32
    - 64
    - 128
    - 256
    - 320
    - 384
2Mrandomread:
  jobname: randread
  mode: randread
  numjobs:
    - 1
  op_size: 2097152
  total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 8
    - 12
    - 16
    - 20
    - 24
    - 28
    - 32
    - 64
    - 128
    - 192
2Msequentialread:
  jobname: seqread
  mode: read
  numjobs:
    - 1
  op_size: 2097152
  total_iodepth:
    - 1

```

```

- 2
- 3
- 4
- 8
- 12
- 16
- 20
- 24
- 28
- 32
- 64
- 128
- 192
32krandomread:
  jobname: randread
  mode: randread
  numjobs:
    - 1
  op_size: 32768
  total_iodepth:
    - 1
    - 2
    - 4
    - 6
    - 8
    - 12
    - 16
    - 24
    - 32
    - 64
    - 128
    - 256
    - 320
    - 384
32ksequentialread:
  jobname: seqread
  mode: read
  numjobs:
    - 1
  op_size: 32768
  total_iodepth:
    - 1
    - 2
    - 4
    - 6
    - 8
    - 12
    - 16
    - 24
    - 32
    - 64
    - 128
    - 256
    - 320
    - 384
384krandomread:
  jobname: randread
  mode: randread
  numjobs:
    - 1
  op_size: 393216
  total_iodepth:
    - 1

```

```

- 2
- 3
- 4
- 5
- 8
- 16
- 24
- 32
- 64
- 128
- 160
- 256
- 512
384ksequentialread:
  jobname: seqread
  mode: read
  numjobs:
    - 1
  op_size: 393216
  total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 5
    - 8
    - 16
    - 24
    - 32
    - 64
    - 128
    - 160
    - 256
    - 512
4Mrandomread:
  jobname: randread
  mode: randread
  numjobs:
    - 1
  op_size: 4194304
  total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 8
    - 12
    - 16
    - 20
    - 24
    - 28
    - 32
    - 64
    - 128
    - 192
4Msequentialread:
  jobname: seqread
  mode: read
  numjobs:
    - 1
  op_size: 4194304
  total_iodepth:
    - 1

```


- 2
- 3
- 4
- 8
- 12
- 16
- 20
- 24
- 28
- 32
- 64
- 128
- 192

4kmixread70:

- jobname: mixread
- mode: randrw
- numjobs:
- 1
- op_size: 4096
- rwmixread: 70
- total_iodepth:
- 1
- 2
- 4
- 6
- 8
- 12
- 16
- 24
- 32
- 64
- 128
- 256
- 320
- 384

4krandomread:

- jobname: randread
- mode: randread
- numjobs:
- 1
- op_size: 4096
- total_iodepth:
- 1
- 2
- 4
- 6
- 8
- 12
- 16
- 24
- 32
- 64
- 128
- 256
- 320
- 384

4ksequentialread:

- jobname: seqread
- mode: read
- numjobs:
- 1
- op_size: 4096
- total_iodepth:

```

- 1
- 2
- 4
- 6
- 8
- 12
- 16
- 24
- 32
- 64
- 128
- 256
- 320
- 384
512krandomread:
  jobname: randread
  mode: randread
  numjobs:
    - 1
  op_size: 524288
  total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 8
    - 16
    - 24
    - 32
    - 40
    - 48
    - 64
    - 128
    - 256
    - 512
512ksequentialread:
  jobname: seqread
  mode: read
  numjobs:
    - 1
  op_size: 524288
  total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 8
    - 16
    - 24
    - 32
    - 40
    - 48
    - 64
    - 128
    - 256
    - 512
64kmixread30:
  jobname: mixread
  mode: randrw
  numjobs:
    - 1
  op_size: 65536
  rwmixread: 30

```

```

total_iodepth:
- 1
- 2
- 4
- 6
- 8
- 12
- 16
- 24
- 32
- 64
- 128
- 256
- 320
- 384
64kmixread70:
  jobname: mixread
  mode: randrw
  numjobs:
  - 1
  op_size: 65536
  rwmixread: 70
  total_iodepth:
  - 1
  - 2
  - 4
  - 6
  - 8
  - 12
  - 16
  - 24
  - 32
  - 64
  - 128
  - 256
  - 320
  - 384
64krandomread:
  jobname: randread
  mode: randread
  numjobs:
  - 1
  op_size: 65536
  total_iodepth:
  - 1
  - 2
  - 4
  - 6
  - 8
  - 12
  - 16
  - 24
  - 32
  - 64
  - 128
  - 256
  - 320
  - 384
64ksequentialread:
  jobname: seqread
  mode: read
  numjobs:
  - 1

```

```

  op_size: 65536
  total_iodepth:
  - 1
  - 2
  - 4
  - 6
  - 8
  - 12
  - 16
  - 24
  - 32
  - 64
  - 128
  - 256
  - 320
  - 384
768krandomread:
  jobname: randread
  mode: randread
  numjobs:
  - 1
  op_size: 786432
  total_iodepth:
  - 1
  - 2
  - 3
  - 4
  - 8
  - 16
  - 24
  - 32
  - 40
  - 48
  - 64
  - 128
  - 256
  - 512
768ksequentialread:
  jobname: seqread
  mode: read
  numjobs:
  - 1
  op_size: 786432
  total_iodepth:
  - 1
  - 2
  - 3
  - 4
  - 5
  - 8
  - 16
  - 24
  - 32
  - 64
  - 128
  - 160
  - 256
  - 512
8krandomread:
  jobname: randread
  mode: randread
  numjobs:
  - 1

```

```

    op_size: 8192
    total_iodepth:
      - 1
      - 2
      - 4
      - 6
      - 8
      - 12
      - 16
      - 24
      - 32
      - 64
      - 128
      - 256
      - 320
      - 384
  8ksequentialread:
    jobname: seqread
    mode: read
    numjobs:
      - 1
    op_size: 8192
    total_iodepth:
      - 1
      - 2
      - 4
      - 6
      - 8
      - 12
      - 16
      - 24
      - 32
      - 64
      - 128
      - 256
      - 320
      - 384
  precondition:
    jobname: precond1rw
    mode: randwrite
    monitor: false
    numjobs:
      - 1
    op_size: 65536
    time: 600
    total_iodepth:
      - 16
cluster:
  archive_dir: /tmp/cbt
  ceph-mgr_cmd: /usr/bin/ceph-mgr
  ceph-mon_cmd: /usr/bin/ceph-mon
  ceph-osd_cmd: /usr/bin/ceph-osd
  ceph-run_cmd: /usr/bin/ceph-run
  ceph_cmd: /usr/bin/ceph
  clients:
    - --- server1 ---
  clusterid: ceph
  conf_file: /cbt/ceph.conf.4x1x1.fs
  fs: xfs
  head: --- server1 ---
  iterations: 1
  mgrs:
    --- server1 ---:

```

```
    a: null
mkfs_opts: -f -i size=2048
mons:
  --- server1 ---:
    a: --- IP Address ---:6789
mount_opts: -o inode64,noatime,logbsize=256k
osds:
- --- server1 ---
osds_per_node: 6
pdsh_ssh_args: -a -x -l%u %h
rados_cmd: /usr/bin/rados
rbd_cmd: /usr/bin/rbd
tmp_dir: /tmp/cbt
use_existing: true
user: cjames
monitoring_profiles:
  collectl:
    args: -c 18 -sCD -i 10 -P -oz -FO --rawtoo --sep ";" -f {collectl_dir}
```